Experiment 3.1

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Subject Name: Numerical Methods **Subject Code: 23SMH-341**

1. **Aim:** To numerically evaluate a definite integral using Simpson's 1/3 Rule and analyze its accuracy for different step sizes.

2. S/W Requirement: Python 3.11

3. Code:

```
def simpson_rule(f, a, b, n):
    if n % 2 != 0:
        print("n must be even")
        return

    h = (b - a) / n
    total = f(a) + f(b)

    for i in range(1, n):
        x = a + i * h
        if i % 2 == 0:
            total += 2 * f(x)
        else:
            total += 4 * f(x)

    return (h / 3) * total
```

```
# NEW FUNCTION
f = lambda x: x**3 + 2*x + 1

a = 1
b = 2
n = 6

result = simpson_rule(f, a, b, n)
print("Integral =", result)
```

4. Output:

```
Integral = 7.75
```

5. Learning Outcome:

- Understand numerical integration of functions.
- Apply Simpson's Rule for estimating definite integrals.
- Analyze effect of step size on accuracy.
- Implement the method using Python.